

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, Distributed Systems Aug 2025 to May 2026

- Automated provisioning, configuration, and **telemetry** for containerized test clusters, cutting experiment setup effort **55%** and making every sweep reproducible from config.
- Validated **14 interoperability scenarios** across networking, dependency, and **failure-recovery** paths to confirm recovery behavior before large experiment runs.
- Built an automated **telemetry pipeline** that collected, normalized, and analyzed experiment data end to end, cutting setup and analysis effort **55%**.

Turito | Software Engineering Intern, Distributed Systems Jun 2022 to Oct 2022

- Slashed p95 CPU **22%** by profiling high-frequency media paths and eliminating excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Cut controlled-disruption failures **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Lowered network egress **14%** and p95 CPU **22%** on a high-frequency event path by removing redundant processing, payload copies, and per-packet logging overhead.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Raised equivalent-load signaling throughput about **31%** by surfacing concurrency bottlenecks that drove targeted backend fixes.
- Cut investigation time about **40%** by building a pipeline that aggregated load-test output from Rust workers into **SQLite** with Python and served comparisons in Flask.

TECHNICAL SKILLS

Languages: Python, Rust, C++17, TypeScript, SQL

Serving & Gateways: OpenAI-compatible gateways, exact and semantic caching, failover, rate limiting, SSE streaming, asyncio, gRPC

Performance: p50/p95 latency profiling, CPU and lock-contention profiling, Rust/Tokio and Locust load testing, throughput tuning

Infrastructure: Docker, Kubernetes, AWS, Linux, CI/CD, telemetry pipelines, structured logging

PROJECTS

Relay, LLM Inference Gateway | *Python, FastAPI, asyncio, SQLite, SSE*

- Cut p50 latency from **759 ms to 44 ms** over a **4,000**-request load test with an OpenAI-compatible gateway doing exact and **semantic caching**, failover, and rate limiting.

Melody Metrics, Big-Data Pipeline | *C++, Hadoop, MapReduce, Hive, Python, XGBoost*

- Processed **1M+** heterogeneous records through C++ **MapReduce** and **Hive** into a curated **140,000**-row dataset powering an **84%**-accurate 81-genre classifier.

Proxima, Vector Search Engine | *C++17, AVX2/SIMD, HNSW, pybind11*

- Reached **0.999 recall** at **12,049 QPS** single-threaded, **1.8x** hnswlib and **2.5x** FAISS, with an HNSW engine written from scratch in C++17.

OPEN SOURCE

- fission/fission:** 3 merged PRs, Kubernetes event and KEDA object capture in support dumps, removed duplicate package enumeration
- google/osv-scalibr:** upstream change to Google's software composition analysis library (open PR, awaiting review)

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Distributed Computing, Computer Networks, Advanced Algorithms, Database Systems, Machine Learning, GPU Programming

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)